

Ping Pong Game

Clone Repo: <https://github.com/codertheashish/Ping-Pong-Game>

1. Introduction

The **Ping Pong Game** is a two-dimensional arcade game developed using **Python** and the **Pygame** library. In this game, the player controls the left paddle using the keyboard, while the right paddle is controlled by an AI opponent. The objective is to hit the ball and score points by making the opponent miss the ball. The first player to reach **7 points** wins the game. The project also includes a **dynamic ball speed system**, **color synchronization**, and a **Game Over screen** with restart and exit options.

2. Purpose

The main purpose of this project is to learn the basics of **2D game development** using Python and Pygame.

This project helps in understanding:

- Game loop
 - Keyboard input handling
 - AI movement
 - Collision detection
 - Score management
 - Ball physics
 - Real-time animation
 - Object drawing using Pygame
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3. Technologies Used

Technology	Purpose
Python	Main programming language
Pygame	Creates the game window, graphics, and handles user input
Random Module	Generates a random starting direction for the ball
Sys Module	Safely closes the game window

4. Main Features

✓ AI Opponent

The right paddle is controlled by a simple AI. It automatically follows the ball and tries to return it. The AI moves slightly slower than the player, making the game balanced and enjoyable.

✓ Dynamic Ball Speed

Each time the ball hits a paddle, its speed increases slightly.

This makes the game more challenging as it continues, while still keeping the movement smooth.

✓ Color Synchronization

Every time the **player** hits the ball:

- The ball changes its color.
- The score text changes to the same color.

The colors cycle in this order:

- White
- Green
- Yellow
- Orange

- Red
- Purple

This feature gives the game a modern and attractive appearance.

✓ Score System

Whenever the ball crosses a player's side:

- The opponent receives **1 point**.
- The ball returns to the center.
- The ball starts moving again in a random direction.

The first player to reach **7 points** wins the match.

✓ Game Over Screen

When either the player or AI reaches **7 points**, a Game Over screen appears.

The player has two options:

- **Press R** → Restart the game.
 - **Press ESC** → Exit the game.
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✓ Keyboard Controls

The game uses simple keyboard controls.

- **Up Arrow** → Move paddle up.
 - **Down Arrow** → Move paddle down.
 - **R** → Restart after Game Over.
 - **ESC** → Exit the game.
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